

Dark Matter

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1 Introduction

Currently, dark matter (DM) represents an outstanding problem in both cosmology and particle physics. Even though the existence of DM is debatable, there are several striking evidences like rotation curve of a galaxy, observation of cosmic microwave background (CMB) radiation, observation of large-scale structure show that about 27% of the mass of our universe is made up of DM. The nature of DM is currently one of the intriguing question to astrophysics, cosmology and particle physics. There are several experiments, particle physics experiments, are dedicated in search of DM particle(s). In the context of particle physics, DM can be made up of one or several new particles which are expected to be electrically neutral, non-baryonic, uncoloured, weakly-interacting with standard model (SM) and stable.

It has been argued that DM can be captured inside the compact star (CS)s like neutron star (NS)s by losing its kinetic energy when the DM particle interact with ordinary matter [Bose:2021yhz]. Over the time, this capture of DM can lead to an accumulation of DM inside CS. Hence it is natural that one can expects the presence of DM in CSs. Further more, DM can be trapped inside CSs during thier formation itself.

In recent years, one of the indirect methods that have gained attention is the study of the DM effects on CSs properties. CSs like NSs or hybrid star (HS)s, (HS is a CS with quark matter in its core), provide an ample of opportunities to study the multidisciplinary physics such as general theory of relativity (GTR), low energy nuclear physics, quantum chromodynamics (QCD) under extreme conditions, etc.

The macroscopic properties of such compact objects like its mass, radius, moment of inertia, tidal deformability in a binary merging system and different modes of oscillations etc. depend crucially on its composition that affects the variation of pressure with energy density or equation of state (EOS). Although these macroscopic properties can be described from the microscopic nature of the matter inside the core of the NS or HS. It's a big question whether the matter inside the compact objects is nuclear matter, or quark matter, or a mixer of nuclear and quark matter. Indeed recent observations (gravtitational wave (GW)s, Neutron Star Interior Composition Explorer (NICER)) have drawn attention on the nuclear interactions at high densities with questions regarding the possible presence of exotic matter in the core. The other observational parameters also possible to unveil the NSs like quasi normal mode (QNM). QNMs in NSs are very much sensitive and directly related to the matter inside the core. As if there is a quark-hadron phase transition in the core of NS, the frequency of the mode enhances. So there might be a possibility of a correlation between QNMs to NS matter properties, like symmetric matter properties, asymmetric matter properties, mass, radius, tidal deformability of NS etc.

Weakly interacting massive particle (WIMP) can accumulate inside NSs due to elastic scattering with nucleons.

From the observation point of view, the details of the internal structure of CSs become important.

The core density of NSs range from $2\rho_0$ to $20\rho_0$ where ρ_0 is the nuclear saturation density. Due to the high density of normal matter inside the core, the collision energy loss of DM particle can be quite significant. This makes the accumulation of DM inside NSs. If the mass fraction of DM reaches a critical value, DM form a self-gravitating dense core supported by degeneracy fermi pressure.

Accumulation of DM in the core of NS can be performed in several ways. The cooling of DM by interacting neutron star matter (NSM) slow down the